Qi Wu

PhD Candidate | Surface Science & Optical Metrology Specialist
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Profile: Materials Scientist and Metrology Expert with 4+ years of R&D experience at imec, a world-leading research hub. Expertise in Surface Modification, CVD processes, and Hardware-Software Integration for customized metrology. Proven ability to bridge fundamental material science with semiconductor manufacturing, specialized in optical sensing and interfacial engineering.

EDUCATION

- **Ph.D. in Chemistry** |2022-Sep. 2026 (expected) KU Leuven & imec, Leuven, Belgium
- **M.Sc. in Electrical Engineering** | 2019-2021 University of Twente, Enschede, the Netherlands
- **B.Eng. in Electrical Engineering** | 2015-2019 Northwestern Polytechnical University, Xi'an, China

TECHNICAL SKILL

- **Surface Engineering:** 2,000+ hours of cleanroom experience. Expert in Self-Assembled Monolayers (SAMs) Chemical Vapor Deposition (CVD) and wet-chemical surface functionalization.
- **Advanced Metrology:** Proficient in ATR-FTIR spectroscopy, Ellipsometry, SEM, AFM, and Dynamic Contact Angle/Wettability analysis.
- **Optical Modeling & Data:** Advanced in optical simulations; Proficient in Python & MATLAB for automated data processing and hardware-software integration.
- **Prototyping:** Experienced in 3D printing and embedded systems (microcontrollers) for building customized laboratory measurement tools.
- **Language:** English (Professional) and Mandarin Chinese (Native)

RESEARCH EXPERIENCE

Ph.D. Researcher | *KU Leuven & imec, Leuven, Belgium (2022 - Present)*

- **Surface Wettability Modification:** Achieved surface hydrophobicity by silane deposition. Investigated the effect of -OH density on the silicon-based surface on the deposition properties. Established a robust process window to ensure high reproducibility of surface energy control.
- **Wettability Characterization of Nanopatterns:** Investigated the wetting properties of nano-trench patterned surfaces, including their geometry dependence and deviations from theoretical models.
- **Custom ATR-FTIR Metrology Tooling:** Designed and built a specialized metrology module for the ATR spectrometer to monitor the wetting progress on patterned surfaces.
- **Anisotropic Wetting Analysis:** Integrated a dual-axis imaging module into conventional contact-angle systems to quantify liquid spreading on asymmetric nano-trenches.
- **Award:** Awarded **Best Student Paper at UCPSS 2025** for innovative surface characterization methodologies.

Graduate Intern | *ARCNL, Amsterdam, the Netherlands (2021)*

- **Mechanical Modeling:** Conducted stress-strain simulations for elastomer microstructure arrays used in low-wear wafer carriers.
- **Industrial Collaboration:** Partnered with cleanroom specialists to optimize micro-component durability in a research-to-manufacturing environment.

INTERESTS

- Tennis, bouldering, hiking, DIY electronics